

The EIRIN Flow-of-funds Behavioural Model of Green Fiscal Policies and Green Sovereign Bonds



Irene Monasterolo^{a,b,*}, Marco Raberto^c

^a Vienna University of Economics and Business (WU), Austria

^b The Frederick S. Pardee Centre for the Study of the Longer Range Future, Boston University (BU), United States

^c DIME - University of Genoa (UNIGE), Italy

ARTICLE INFO

Keywords:

Flow-of-funds
Stock-flow consistency
Green fiscal policies
Green sovereign bonds
Endogenous money
Distributive effects

ABSTRACT

Fiscal and monetary policies, as well as new financial instruments, could play a key role to meet the Paris Agreement. However, deep uncertainty characterizes their design and their potential effects on growth, financial and credit market stability, and inequality. We develop the EIRIN flow-of-funds behavioural model to simulate the introduction of green fiscal policies and green sovereign bonds, and we display their effects on firms' investments in the brown and green sector, on unemployment, on the credit and bonds market. EIRIN is Stock-Flow Consistent and is rooted on a balance sheet approach. It adopts a Leontief production function with no substitution of the production factors, i.e., Labour, Capital, and Raw Materials. Its sectors are endowed with adaptive behaviours and expectations, and interact with the others and the foreign sector through a set of markets. Simulations show that green public policies can promote green growth by influencing firms' expectations and the credit market. Green sovereign bonds represent a short-term win-win solution, while green fiscal measures have higher immediate distributive effects that induce negative feedbacks on the economy. These results are influenced by the conditions (fiscal, budgetary and public debt/GDP) in which both measures are implemented.

1. Introduction

Mature economies such as the European Union (EU) are still struggling to get out of the so-called “secular stagnation”. At the same time, climate change was recently identified as an additional source of risk for financial markets and for the real economy (see for instance Carney, 2015; ESRB, 2016; Batten et al., 2016). Besides the well-known climate physical risks (Stocker et al., 2013), recently climate transition risks started to be investigated, in particular those related to carbon stranded assets, i.e., assets that are at risk of losing much of their value as a result of unburnable reserves of fossil fuels (McGlade and Ekins, 2015). The realization of carbon stranded assets is expected to increase price volatility of both carbon-intense and renewable energy assets, affecting negatively the former and positively the latter (Fischer, 2015; Lazarus and Tempest, 2014). The reason is that the introduction of market-based solutions to climate change (such as a global carbon tax) aimed at decarbonizing the economy could directly and immediately affect the revenues and thus the assets' value of companies in carbon-intense sectors, and as a consequence the value of the portfolios of investors exposed to them.

Risk transmission from climate change to the financial sector started to be analysed and appears to be substantial (Dietz et al., 2016), with potential systemic ramifications and cascade effects throughout the entire financial network (Battiston et al., 2017). However, capital is flowing in the low-carbon economy at a much slower pace than needed to meet the 2°C target (Volz, 2017). While current investments in renewable energy reached USD 242 billion (bn) in 2016 (BNEF/UNEP, 2017), the International Energy Agency (IEA) has recently estimated that the retrofitting of the energy sector by 2035 would require investments worthy USD 53 trillion (trn).

Better disclosure of information on climate-related financial risk from the one hand (FSB TFCD, 2017), and the introduction of a stable green policy framework from the other hand (Stern, 2016) are recommended to provide investors the right signals and incentives to invest in a sustainable, inclusive and innovation-based growth. In this context, the role of green policies such as green fiscal and green monetary policies (see for instance Monnin and Barkawi, 2015; Mazzucato and Penna, 2015; Campiglio, 2016), and the introduction of new financial instruments, such as green sovereign bonds, gained attention among academics and practitioners. The political feasibility and

* Corresponding author at: Welthandelsplatz 1/D5, 1020 Vienna, Austria.

E-mail addresses: irene.monasterolo@wu.ac.at (I. Monasterolo), marco.raberto@unige.it (M. Raberto).

the effectiveness of green public policies started to be addressed as well (Rozenberg et al., 2013; Rozenberg et al., 2017; Nemet et al., 2017). However, there is high uncertainty about their design, due to the lack of consolidated knowledge on their direct and indirect effects on the real economy and on financial markets. Moreover, their distributive effects and trade-offs, in terms of income inequality and wealth concentration across economic sectors and social groups, has not been properly addressed yet. Modelling methods based on general or partial equilibrium approaches, such as the Integrated Assessment Models (IAMs), are not able, by construction, to represent a complex system where the presence of cross-sector feedback loops and time delays at the macro-economic level, and heterogeneous short-term thinking agents at the micro-economic level, determines non-linearity and policy uncertainty (Mercure et al., 2016; Balint et al., 2017; Ackerman et al., 2009; Ackerman and Munitz, 2016; Scricciu et al., 2013). Most important, IAMs don't include a credit and financial sector and omit modern money theory and banks' endogenous money creation (Wray, 2015; McLeay et al., 2014). This means that they are not able to display neither the dynamics of private debt (including the implications on risk creation and diffusion from the credit market to the real economy), nor the role of Central Banks on investors' expectations. Therefore, scholars started to recognize the need for bottom-up and out-of-equilibrium models rooted on complex system science to understand sources of systemic risk emerging from the interaction between climate change, the real economy, the credit and financial markets (Farmer et al., 2015; Battiston et al., 2016b). In particular, Rezai and Stiglitz (2016) called for the development of a new generation of models in ecological macro-economics able to integrate the micro-foundations of the models with a meso and macroeconomic level of analysis to better understand the feedback loops between the ecosystem, the real economy and the financial sector.

With the aim to contribute to this stream of research, we introduce EIRIN, which is a Stock-Flow Consistent (SFC) model rooted on a neo-Schumpeterian, evolutionary economics approach. EIRIN features heterogeneous economic sectors and subsectors characterized by adaptive behaviours and expectations (households, firms), heterogeneous capital goods characterized by different resource intensity, a credit sector characterized by endogenous money creation, and a foreign sector. In addition, EIRIN connects these elements with policy agents, such as a government that decides on the fiscal policy and issues green financial products (i.e., green sovereign bonds), and a Central Bank in charge of setting the monetary policy. EIRIN includes some novel elements to the expanding field of the ecological macroeconomics and environmental economics literature (see Dafermos et al., 2017; Lamperti et al., 2015; Ponta et al., 2016; Bovari et al., 2017). First, it endogenizes green technology investments and displays their effects on the changes in green technology adoption and thus on the level of resource efficiency of the production process, on the structure of the real economy, on credit market performance and on income distribution. This solution is alternative to the conventional environmental economic models that adopt a cost-benefit approach through market-based pricing (Stern, 2006; Weitzman, 2009), and are well-known for underestimating the negative externalities of climate change, as displayed by the social cost of carbon, and the distributive effects (see Ackerman and Stanton, 2012; Pindyck, 2013).

Second, it analyses the effects of resource intensive production and consumption on the performance of the real economy (e.g., employment, capital accumulation), of the balance of payments and credit market.

Third, it simulates two different sets of green public policies – i.e. green sovereign bonds *vis a vis* green fiscal measures – through which the government covers the cost of the introduction of green subsidies. Green sovereign bonds are issued by the government and subsidize firms' green investments, thus they have a clear conditionality associated with their use. Green public policies influence firms' green/brown investment choices, macro-economic performance and credit

conditions. In addition, the Central Bank influences investment and consumption's decisions of the economic agents by setting the nominal interest rate. These modelling solutions are important when we want to understand the channels of transmission of different sets of green public policies on the sectors and subsectors of the economy.

The name EIRIN, which in ancient Greek means “harmony, peace”, was chosen because we believe that the goals of economic development and sustainability are not mutually exclusive but could be instead mutually reinforcing under specific policy conditions.

2. Methodology

2.1. Model Outline

EIRIN is a demand-driven model in the (post-) Keynesian tradition. EIRIN is SFC (Godley and Lavoie, 2007; Lavoie, 2014; Caverzasi and Godin, 2013; Caiani et al., 2015), and adopts a double-entry balance sheet accounting approach (Raberto et al., 2012; Bezemer, 2012) that contributes to increase the transparency and the consistency of results by tracking all the transactions within the economy, by recording all the changes in the stocks of assets and liabilities for each economic sector, and by displaying their relation with the changes in the flows across sectors. In so doing, we assure the correspondence between assets and liabilities, whose difference equals to zero. EIRIN can be described as a flow-of-funds model in so far it captures all the financial transactions and financial positions of all the sectors in the economy¹ in terms of production, consumption, employment, income inequality, the credit and the bonds market.

The EIRIN economy is shaped on a middle-high income country (such as the member states of the European Union, or the United States), which relies on fossil fuels and raw materials for its production and consumption processes. EIRIN is characterized by deterministic dynamics, and by a structure based on stocks interconnected through flows, where the stocks represent the balance sheet entries of the sectors (or representative agents), and the flows represent their cash and material in-and-out flows. EIRIN's elementary units and building blocks are the heterogeneous sectors of the economy, which could be considered as representative agents characterized by own adaptive behaviours and expectations. Indeed, in addition to stocks-and-flows' balance sheet relations, and to a dynamics based on the stocks' rates of change, EIRIN's sectors display an adaptive decision making and are endowed with own behavioural rules. Therefore, EIRIN's sectors' specific behavioural characteristics make them closer to the agents of an Agent-based model (ABM) than to the sectors of a System Dynamics model (SD), which displays aggregate sector behaviour. In this way, the model can support the representation of endogenous decision-making, which is fundamental to understand the drivers of sectors' intertemporal behaviours and their consequences on the dynamics of the system. This is the case, for instance, of investment decisions made by the consumption goods producers (i.e., the Net Present Value (NPV) of brown versus green investment decisions), who compare the short-term costs of investments with the discounted value of future expected cash flows.

EIRIN is populated by the following sectors: a household sector, which is divided into a worker and a capitalist agent; a banking sector, which is represented by a commercial bank; a Central Bank; a consumption goods production sector, represented by a firm (CGP); a capital goods production (KGP) sector, which is divided into two agents (KGP brown and KGP green); a government; a foreign sector, which provides the raw materials imported in the economy. The sectors interact through the following set of markets: consumption goods, capital

¹ The term flow-of-funds is widely used in post-Keynesian literature (see Tymoigne, 2006) in the context of macro-economic models based on systems of interrelated balance sheets, as well as by major Central Banks in the world. See for instance the US Federal Reserve, <https://www.federalreserve.gov/releases/Z1/Current/> and the European Central Bank, Bè Duc and Le Breton, 2009.

goods, labour, raw materials, credit market and green/brown bonds market.

In the model, the government is responsible for setting the fiscal policy and for collecting tax revenues. In addition, the government is subject to public deficit targets (e.g., the Maastricht criteria in the EU). The Central Bank sets interest rates, the inflation and the employment targets according to a Taylor-like rule. The production function of the consumption goods sector is based on a Leontief (1941) with no substitution of the production factors, i.e. labour, capital, and raw materials (these latter include fossil fuels).

Prices are set by the supply side, i.e., the consumption and capital goods sectors, and are based on mark-up, set at 10% on unit's costs.

2.2. Model Distinctive Features

In comparison with recent models that adopt a SFC approach and are applied to the analysis of the growth – environment – policy relation (see for example Jackson and Victor, 2015; Naqvi, 2015), EIRIN introduces the following novelties in terms of model's structural heterogeneity and behavioural foundations.

2.2.1. “Brown” and “Green” Capital Goods Sectors

Capital goods set the consumption goods sector's production capacity and technology endowment, and reflect different levels of resource efficiency. By building on Acemoglu et al. (2014) and Lamperti et al. (2015), we introduce a “brown” and a “green” capital goods sector and corresponding producers, where the green capital goods allow for a more resource-efficient production process than the brown ones. However, we don't embrace a dichotomous structure but we allow the consumption good producer to choose to gradually divest from brown, and to implement “shades of green” in its investment decisions, according to how it discounts the future (as displayed by the NPV). For the sake of simplicity, we assume that both the brown and the green sector uses only labour as a production factor for the production of capital goods. This modelling choice helps us to focus on the differences between the brown and the green sector in terms of effects on the macroeconomic performance and on the credit market, and on the role of public incentives as a driver of the low-carbon transition. We assume that the brown and the green sectors differ for their level of resource efficiency, technological endowments and human capital. The green sector is characterized by high levels of technology and human capital (i.e., labour force's skills) that allow the production to be resource efficient, with a minor contribution to resource depletion and to CO2 emissions, in line with the consolidated literature on eco-innovation based technology (Quadrio Curzio et al., 2012; Costantini and Mazzanti, 2012; Consoli et al., 2016). For this reason, green capital goods are more costly than brown capital goods, being equal their productive capacity, because they require more R&D and human capital endowments. Resource efficiency is assumed to be higher in the green sector, where it tends to infinite. This rather strict assumption allows us to abstract from deeper technological considerations on resource efficiency, and thus to keep the model structure simple enough so to allow the reader to concentrate on the real focus of our simulation exercise, i.e., the role of green public incentives on promoting firms' green investments.

2.2.2. Endogenous Investment Decisions by the Consumption Goods Sector

The CGP agent decides whether to invest in green or brown capital goods according to the NPV. We opted for the NPV because it is widely used in business planning and because it allows to display agents' intertemporal investment behaviour, by comparing short term costs with long term benefits. In particular, we explicit the role of raw materials' costs in the NPV decisions through the introduction of a resource intensity parameter to model raw materials' cost on the CGP' investments decisions. The value of the parameter depends on the raw materials' intensity of the production process and, by construction, is higher in the

brown sector.

2.2.3. Resource use by the Consumption Goods Sector

By building on the intuitions of Daly (1995), and Kemp-Benedict (2014), and consistently with post-Keynesian pricing (e.g., Coutts and Norman, 2013), we consider the value of output as a multiple of raw materials' costs and wages. In so doing, we explicit the role of raw materials in the production function.

EIRIN is modelled on the case of a domestic economy, which is dependent on the foreign sector for the import of raw materials, which are in turn needed by the brown production sector. Raw materials' price is exogenously given and it is set as a constant in our computational experiments. We assume the irreplaceability of raw materials in the production function given the nature of technology and lack of substitution alternatives in the short-term, which is the focus of model's simulations. This means that the CGP that invests in green would still rely in minimal part on brown capital goods at the beginning of the transition process, this part being gradually phased-out with the depreciation of brown capital and by the increase of green capital goods. The demand for raw materials depends on the production technology of the CGP, i.e., whether its capital endowment is composed predominantly by brown or green capital goods. The CGP that opts for brown production technology faces higher production costs because the brown production is more resource intensive than the green one, and thus needs to import raw materials to maintain the production at the desired level. At the opposite, the use of green capital goods by the CGP contributes to increase the level of resilience by using less overall raw materials and natural resources to obtain the same output level, thanks to the higher levels of technology and human capital employed. Capital depreciation rate is set at 1% per time step.

2.2.4. Household Sector

The household sector is divided into two income classes, i.e., a worker and a capitalist (see Goodwin, 1967 and the Lotka-Volterra's predator-prey model). This solution allows to display the distributive effects of public policies and market-driven forces on wealth concentration and inequality. The household sector includes (i) a worker agent, representative of the working class, whose income is made by salary and (ii) a capitalist agent, representative of the capitalist class, whose wealth is represented by government bonds and the equity shares of KGP, CGP and the banking sectors. It follows that the capitalist agent earns the whole capital income (bonds' coupons and dividends) of the economy. Both the worker and capitalist representative agents make consumption plans based on the buffer-stock theory of savings (Deaton, 1991; Carroll, 2001).

2.2.5. Labour Market

We assume that the representative worker agent offers a constant labour supply (N_{tot}) characterized by heterogeneous skills. Skills are uniformly distributed in a continuous range between the lowest value (zero) and the highest value, normalized to 1. The fraction of the labour supply employed in the green sector (i.e., the green KGP) includes all the skills in the upper range, while the fraction employed in the brown sector (i.e. the brown KGP and the CGP) is characterized by skills in the immediately lower range. For instance, if the green sector employs a fraction N_{green}/N_{tot} of labour supply, then the skills of the employed labour force share range in the interval $(1-x, 1]$, where $x = N_{green}/N_{tot}$ whereas if the brown sector employs a fraction N_{brown}/N_{tot} of the labour supply, the skills range in the interval $(1 - (N_{green} + N_{brown})/N_{tot}, 1 - N_{green}/N_{tot}]$. The least skilled fraction of labour supply, i.e. $(0, 1 - (N_{green} + N_{brown})/N_{tot}]$ is unemployed. Wages in both sectors are set according to the skills levels, which are higher in the green sector, relatively to the average wage, which only depends on the general employment level. The details about wages' dependency on skills are provided in Section 2.5.

Table 1
Sector balance sheet matrix according to the Godley-Lavoie framework. Uppercase letters are used to indicate the stocks; subscripts represent the index of the sector/representative agent to which the stock refers. The Foreign Sector (RoW) column displays the cumulated reserves of the foreign sector resulting from the domestic economy's import of raw materials.

	Households (H)	Worker (Hc)	Capitalist (Hk)	CGP (C)	KGP (Kgreen/ Kbrown)	Bank (B)	Government (G)	Central Bank (CB)	Foreign sector (ROW)	Σ
Tangible capital				$+p_{K_{green}}K_{green} + p_{K_{brown}}K_{brown}$						$+p_{K_{green}}K_{green} + p_{K_{brown}}K_{brown}$
Inventories				$p_c I_c$						$p_c I_c$
Government bonds ($\eta_{brown}, \eta_{green}$)			$+ \eta_{brown} p_{bond}$			$+ \eta_{green} p_{bond}$	$-(\eta_{green} p_{bond} + \eta_{brown} p_{bond})$			0
Debt/credit				$- Loans$		$+ Loans$		$+ Loans_{CB}$		0
HH, CGP, KGP, Gov deposits		$+ M_{Hw}$	$+ M_{Hk}$	$+ M_C$	$+ M_k$	$- Loans_{CB}$	$+ M_G$			0
Domestic reserves						$- Deposits$		$- Deposits_{CB}$		0
ROW reserves						$+ M_B$		$- ForeignLiab_{CB}$	$+ M_{Row}$	0
Gold in the vault*								$+ M_{CB}$		0
Equity shares/			$+ CGP \text{ shares value}$	$- E_C$						0
			$+ KGP \text{ shares value}$		$- E_K$					0
			$+ Bank \text{ shares value}$			$- E_B$				0
Equity (net worth)	$- E_{Hw}$	0	$- E_{Hk}$	0	0	0	$- E_G$	$- E_{CB}$	$- E_{Row}$	$- E_{Hw} - E_{Hk} - E_G - E_{CB} - E_{Row}$
Σ	0	0	0	0	0	0	0	0	0	0

Table notation:

$p_{K_{green}}/p_{K_{brown}}$: price of green/brown capital goods; K_{green}/K_{brown} : capital goods endowment (green/brown).

p_c : consumption goods price; I_c : CGP's inventory quantity.

$\eta_{green}/\eta_{brown}$: amount of green/brown government bonds, p_{bond} : bond price).

The gold in the vault is reported in monetary value.

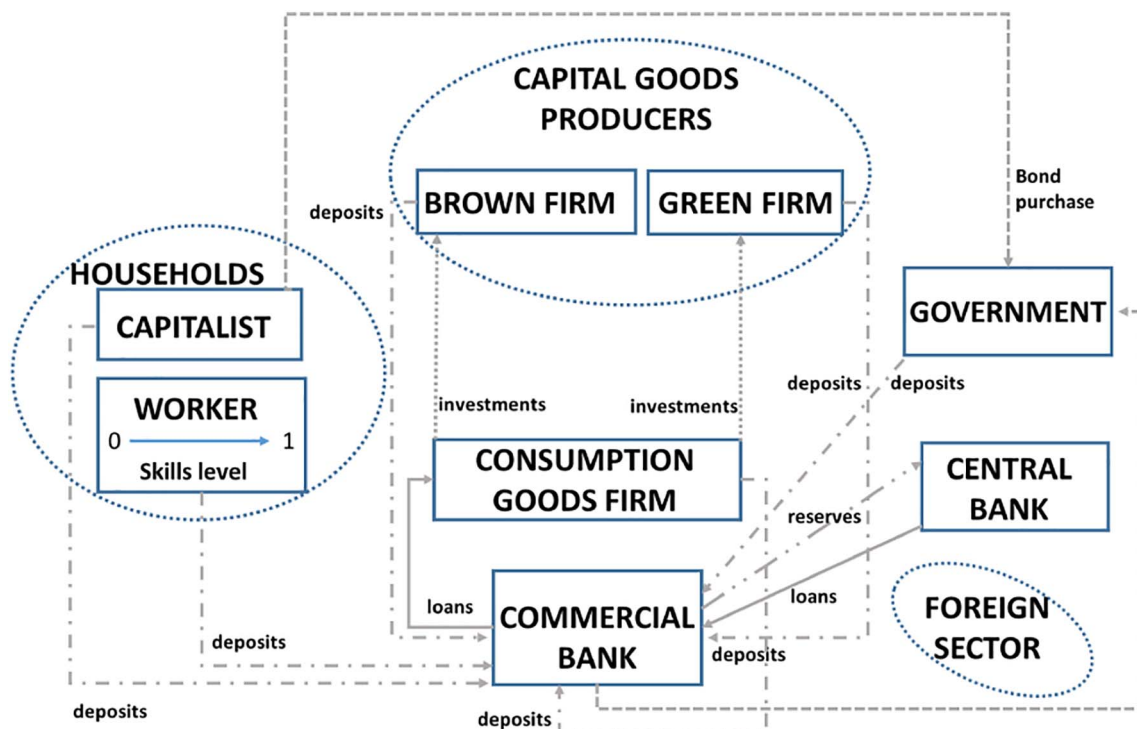


Fig. 1. Capital account flows of the EIRIN economy.

2.3. Model Structure

In Table 1, following the framework introduced by Godley and Lavoie (2007), we present the balance sheet matrix of the model, which displays the balance sheet of the representative agents/sectors and their interdependencies. Each column represents the balance of a representative agent/sector. Assets have a positive sign while liabilities have a negative sign. Each column always sums to zero to highlight the definition of equity (or net worth). The table rows also sum to zero in most cases to show that what is an asset for an agent is a liability for another agent.

As a difference from the balance sheet matrix, which displays the capital accounts balance sheet entries (stocks), Figs. 1 and 2 report all EIRIN's flows in capital and current accounts. Fig. 1 displays the capital account flows. In particular, the dashed lines represent the money flows due to the brown sovereign bonds purchase by the capitalist household, and due to the green sovereign bonds bought by the commercial bank. The dotted lines represent the money flows due to investments. The thick lines display the commercial bank's loans to the consumption goods firm, and the Central Bank's loans to the commercial bank (in case the latter is short of liquid assets). The remaining lines display the deposits from all economic actors to the commercial bank, and from the commercial bank to the Central Bank (reserves).

Fig. 2, instead, displays the current account flows. Each arrow is oriented according to the money flows that it represents in the model. Capital and consumption goods producers pay the commercial bank an interest on debt, pay dividends to the capitalist household, and pay wages to the labour force. The commercial bank pays interests on loans to the Central Bank for borrowing in case of need of liquid assets, and pays dividends to the capitalist household. The government pays interests on green bonds bought by the commercial bank, and on brown bonds bought by the capitalist household. The government also provides subsidies to the consumption goods firm for green investments. The consumption goods firm imports raw materials from the foreign sector for the brown production. All the actors pay taxes to the government.

2.4. Model Timing of Events

The following events occur sequentially at each simulation step:

1. **Monetary policy decided by the Central Bank.** The Central Bank sets the interest rate² according to a Taylor-like rule. The interest rate depends on the inflation and output gap, which is measured as an employment gap, i.e. the distance to a target level of employment.
2. **Fiscal policy designed and implemented by the government.** The government adjusts the level of taxation on labour and capital income in order to reach a zero deficit target. This level is consistent with the recent budget balanced provisions adopted in the EU, following the so-called Fiscal Compact Treaty³ and embodied in many national Constitutions, such as in the German⁴ and Italian⁵ ones.
3. **Money wage setting.** The new money wage (i.e., nominal wage) is determined according to a Phillips curve-like rule, as in Keen (2013), where the percentage change in the nominal wage depends on the level of unemployment recorded in the previous time step.
4. **Consumption goods market.** The worker and capitalist households set their nominal demand for consumption goods. The CGP provides real supply based on the available inputs⁶ (i.e. the number of employees, capital endowment) and sets prices as a fixed mark-up μ_C on unit costs. The transaction takes place. Finally, based on its recent sales and its inventory level, the CGP makes its production plan for the near future, i.e. the next simulation step.
5. **Green/brown capital goods.** The new green/brown capital goods ordered during the previous simulation step are delivered to enrich

² It is worth noting that the Central Bank's rate determines banks' loan rate r_D that plays a relevant role in the model because it is used as discount rate in the net present value calculations for firm investments' decisions.

³ http://europa.eu/rapid/press-release_DOC-12-2_en.htm.

⁴ <https://www.btg-bestellservice.de/pdf/80201000.pdf> (see art. 109 paragraph 3).

⁵ http://www.quirinale.it/qrnw/costituzione/pdf/costituzione_inglese.pdf (see art 81).

⁶ Raw materials are a production input. However, we assumed that the consumption goods sector is never rationed in the demand for raw materials, which is provided in infinite supply by the foreign sector (ROW) at a given (constant) price and delivered simultaneously to meet the production needs.

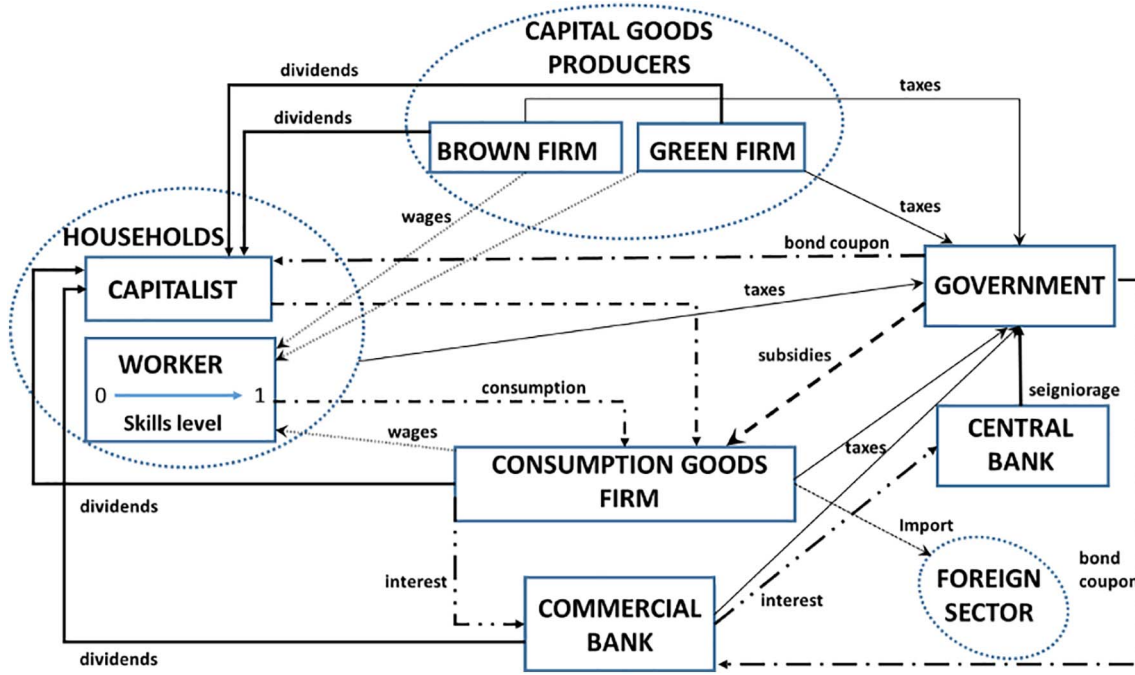


Fig. 2. Current account flows of the EIRIN economy.

the capital endowment of the CGP. Capital goods will be ready to be used for the production of consumption goods at the next simulation step. Green/brown capital goods production firms set the new prices of future production of green/brown capital goods as a fixed mark-up on unit costs (represented only by wages).

6. **Investment demand.** The CGP sets the investment demand for new green/brown capital goods according to the **expected future sales** and the present value of expected additional future cash flows due to investments, net their present costs (i.e., the NPV).
7. **Credit market.** We assume that investments are always entirely financed by credit, which is provided **by the commercial bank**. However, the commercial bank agent is subject to a capital adequacy requirement (CAR) constraint, consistently with the Basel II requirements. Thus, credit may be rationed to the CGP due to insufficient equity capital on the commercial bank's side. In case of capital rationing, the CGP revises its investments downward.
8. **Labour market.** The production sectors set labour demand, and the labour force is reallocated among them. Demand for labour from the CGP and KGP is set according to **their production plans** and their **production technology** (Leontief technology). Labour supply is represented by the worker agent and is inelastic to wages. The **green capital goods production sector has priority on labour supply over the brown one**, which in turn has priority on the consumption goods production sector.
9. **Production of green/brown capital goods.** Green/brown capital goods sectors produce to meet the request of the CGP. The demanded investments will be delivered to the CGP at the next simulation step.
10. **Accounting.** CGP and KGP account sales proceed and input costs, compute their income statements and, **if the debt to equity ratio is lower than a given threshold, they pay-out all the profits** (if positive) as dividends to the capitalist household. Otherwise, their profits contribute to increase their equity capital.⁷ The commercial

bank accounts interest revenues and costs (in the case of loans from the Central Bank) and pays out the whole profits as dividends to the capitalist household, except in the case when the demand for credit is higher than its credit supply capacity, which is set by the capital adequacy rule.

2.5. Model's Behavioural Equations

The behavioural equations related to the timing of events described in 2.4 are provided here.

2.5.1. Monetary Policy Decision by the Central Bank

The Central Bank's interest rate r_{CB} is set according to a Taylor-like rule as follows:

$$r_{CB} = \omega_{\pi} (\pi - \bar{\pi}) + \omega_u (\bar{u} - u) \quad (1)$$

where π and u are endogenous variables of the model.

In particular, π is the **inflation of consumption goods** price p_C , computed as $\pi = \Delta p_C / p_C$ between two consecutive simulation steps; u is the **unemployment rate**, computed as $u = 1 - \frac{N_{brown} + N_{green}}{N_{tot}}$, where $N_{brown} = N_{K_{brown}} + N_C$ is the share of the labour force N_{tot} employed in the brown sector, i.e., the CGP and the brown KGP.

$N_{green} = N_{K_{green}}$ is the share employed in the green sector, i.e., the KGP green (Kgreen). N_{tot} is the fixed size of the labour force provided by the worker agent. We name the ratio $\frac{N_{brown} + N_{green}}{N_{tot}}$ as the employment rate e .

Furthermore, four parameters can be identified: ω_{π} (weight of the inflation term), ω_u (weight of the unemployment term), $\bar{\pi}$ (inflation target) and $\bar{u}$ (unemployment target).

2.5.2. Government's Fiscal Policy Decision

The government computes its budget balance as the difference between tax revenues and government's expenses, which are represented by (i) the green subsidies G paid to the consumption goods firm, and (ii) the coupons paid on the outstanding government (brown and green) bonds, i.e. $(n_{green} + n_{brown})c_{bond}$. Then, **the government adaptively adjusts the tax rate τ , which is applied to the labour income** (wage) of the worker agent and to **the capital income** (dividends) of the capitalist, to **meet the zero target level of the budget balance**. If the budget balance is

⁷ Earnings are usually paid out as dividends to the capitalist to guarantee a regular capital income stream, and keep its contribution to the demand of consumption goods regular over time. Earnings are retained by the consumption goods producer only to the extent that is necessary to keep the financial leverage (debt to equity ratio) under a given threshold.

negative (budget deficit), the tax rate is increased by a fixed amount $\Delta\tau$. If the budget is positive and higher than a given threshold, then the tax rate is decreased by the same fixed amount $\Delta\tau$. Otherwise, the tax rate τ is kept constant.

Furthermore, if the government's deposits are lower than a given positive threshold $\overline{M}$, i.e., $M_G < \overline{M}$, the government issues a new amount Δn_{brown} of (brown) bonds to cover the gap:

$$\Delta n_{brown} = (\overline{M} - M_G)/p_{bond} \quad (2)$$

where p_{bond} is the government (brown/green) bond price. All newly-issued brown bonds are bought by the capitalist household.

Finally, in the “Green Incentive/Bonds” scenario, green subsidies expenses G are not accounted in the budget balance but are financed directly through the issuance of green sovereign bonds for an amount Δn_{green} that, given the price p_{bond} , exactly covers the green subsidies expenses, i.e.,

$$\Delta n_{green} = G/p_{bond} \quad (3)$$

All newly-issued green bonds are bought by the only institutional investor of the EIRIN economy, i.e. the commercial bank, and represent an eligible asset that could be sold to the Central Bank in case of green quantitative easing (QE). The distinction between the buyers of green and brown bonds allows us to represent in a simple way that green sovereign bonds, being innovative financial instruments, are potentially associated to a higher risk. Therefore, they are first bought by institutional investors,⁸ while the capitalist household would focus on traditional brown bonds instruments.

Both green and brown bonds are infinitely-lived assets and pay a fixed coupon c_{bond} at every simulation step. It is worth noting that we stipulate that the bond price p_{bond} , which is the same for both green and brown bonds, varies through time, being defined as the present value of the infinite stream of coupons:

$$p_{bond} = \frac{c_{bond}}{(1 + \mu_{bond})r_{CB}} \quad (4)$$

where the parameter μ_{bond} shall be considered as a fixed risk premium over the Central Bank rate that discounts the government bond coupon's stream.

2.5.3. Money Wage Setting

The average money wage w is subject to a variation Δw between two consecutive simulation steps given by:

$$\Delta w = (-\gamma_1 + \gamma_2 e) w \quad (5)$$

where γ_1, γ_2 are positive parameters subject to the condition $\gamma_2 > \gamma_1$. The rationale of the above formula is to set the money wage growth rate $\Delta w/w$ between the negative value $-\gamma_1$, in the case labour force is completely unemployed, i.e., $e = 0$, and the positive value $-\gamma_1 + \gamma_2$, in the case of full employment of the labour force, i.e., $e = 1$. It is worth noting that the money wage steady state given by $e = \gamma_1/\gamma_2$ leaves the money wage unchanged.

Given the average money wage w , wages in the brown and in green sector are set according to the following equations:

$$w_{brown} = ((1 - z)w_{max} + zw_{min} + w_{min})/2 \quad (6)$$

$$w_{green} = ((1 - z)w_{max} + zw_{min} + w_{max})/2 \quad (7)$$

where $z = N_{green}/(N_{brown} + N_{green})$ is the share of the labour force in the green sector out of total employed labour force, w_{min} is the minimum wage, say a legally set fixed lower bound for the wage dynamics, and w_{max} is the maximum allowed wage, determined by w_{min} and w as $w_{max} = 2w - w_{min}$.

It is worth noting that the two wage setting equations for both the

green and the brown sectors have been determined consistently with the model design, and are characterized by the following properties:

1. $w_{green} > w_{brown}$ for any value of z ;
2. w_{brown} coincides with the average wage w in case of no labour force employed in the green sector ($z = 0$);
3. w_{green} coincides with the average wage w in case of no labour force employed in the brown sector ($z = 1$);
4. the wage bill $Y_w = N_{brown}w_{brown} + N_{green}w_{green}$ of the economy, consistently with the setting mechanism for the average wage, does not depend on the labour force allocation in the brown/green sectors but only on its employment level e . In this respect, it is possible to prove the identity:

$$N_{brown}w_{brown} + N_{green}w_{green} = (N_{green} + N_{brown})w \quad (8)$$

2.5.4. Consumption Goods Market

The representative worker sets its nominal demand for consumption C_{Hw} as:

$$C_{Hw} = Y_w^{net} + \rho(M_{Hw} - \varphi Y_w^{net}) \quad (9)$$

where: Y_w^{net} is the net labour income of workers, computed as $Y_w^{net} = (1 - \tau)Y_w$, given the tax rate τ , ρ is a parameter that determines the speed of adjustment of consumption, and φ is a parameter that sets the target level of liquid assets M_{Hw} to net income Y_w^{net} . The rationale of the buffer stock theory of consumption (Deaton, 1991; Carroll, 2001) here used is that a consumer adjusts his/her consumption path around his/her net income, considering a target level of liquid wealth to income ratio. Therefore, the consumer spends more (less) than his/her net income if his/her actual liquid wealth to income ratio M_{Hw}/Y_w^{net} is higher (lower) than the target level φ .

The capitalist household sets the nominal demand for consumption C_{Hk} according to the same framework:

$$C_{Hk} = Y_k^{net} + \rho(M_{Hk} - \varphi Y_k^{net}) \quad (10)$$

$Y_k^{net} = (1 - \tau)Y_k$, is the net capital income, where the gross capital income Y_k is given by brown bonds' coupons and the sum of all dividends paid by the CGP, d_C , the commercial bank d_B , the KGP brown and green, $d_{K_{brown}}$ and $d_{K_{green}}$, i.e.

$$Y_k = n_{brown}c_{bond} + d_C + d_B + d_{K_{brown}} + d_{K_{green}} \quad (11)$$

Y_k^{net} is then the capital income net of the taxes levied by the government at the rate τ .

On the supply side, the consumption goods firm produces an amount q_C of consumption goods according to a Leontief production technology, characterized by no substitutability among production factors, i.e.,

$$q_C = \min(\gamma_N N_C, \gamma_K K_{tot}, \gamma_R q_R) \quad (12)$$

where N_C is the labour force employed in the consumption goods production sector, K_{tot} is the total capital endowment with productivity γ_K (a fixed parameter) and q_R is the raw material amount required. Green and brown capital goods are perfectly substitutable as regards production capacity, thus the total capital endowment K_{tot} of the CGP is given by:

$$K_{tot} = K_{brown} + K_{green} \quad (13)$$

where: K_{brown} and K_{green} are the endowment of both brown and green capital goods, respectively. The variables γ_N and γ_R represent labour productivity (whose growth rate is exogenously fixed as ξ_N) and resource efficiency, respectively. The resource efficiency variable γ_R is set dynamically according to the share of green and brown capital goods over total capital endowment and their resource intensity parameters, namely a_{brown} and a_{green} , which give the amount of raw material per unit of output. In particular, we define an average resource intensity variable depending on the composition (green/brown) of capital goods as:

⁸ This claim is confirmed by the results of the Climate Bonds Initiative State of the Market report (2016).

$$\alpha = \alpha_{\text{green}} \frac{K_{\text{green}}}{K_{\text{tot}}} + \alpha_{\text{brown}} \frac{K_{\text{brown}}}{K_{\text{tot}}} \quad (14)$$

and then we define γ_R as: $\gamma_R = 1/\alpha$.

Unit production costs u_C are then given by: $u_C = \alpha p_R + \frac{w_C N_C + r_D \text{Loans}}{q_C}$, where p_R is the price of raw materials, w_C is the wage offered by the consumption goods firm, assumed equal to the wage in the brown sector, i.e. $w_C = w_{\text{brown}}$ and $r_D \text{Loans}$ is the interest bill paid on loans at rate r_D . The consumption goods price p_C is then set as a fixed mark-up μ_C on unit costs, i.e. $p_C = (1 + \mu_C)u_C$.

Transactions of consumption goods take place for an aggregate amount q set at the minimum between real demand and real supply, i.e. $q = \min(I_C + q_C, \frac{C_{\text{Hw}} + C_{\text{Hk}}}{p_C})$, where the supply of goods takes also into account the consumption goods firm's inventory I_C . If demand is higher than supply, both the capitalist and the worker are rationed proportionally to their demand. The share of newly produced goods q_C that is eventually unsold is accumulated in the consumption goods firm's inventory, I_C .

Finally, based on recent sales and inventory level, the CGP makes its production plan $\hat{q}_C$ for next simulation step.

2.5.5. Green/Brown Capital Goods Pricing

Units' costs of both KGP green and brown are given by their unit labour costs, as labour is their only production factor. The price $p_{K_{\text{green}}}$ and $p_{K_{\text{brown}}}$ of green and brown capital goods is then set as a fixed mark-up μ_K on units labour costs as follows:

$$p_{K_j} = (1 + \mu_K) \frac{w_j}{\gamma_N} \quad (j = \text{green/brown}) \quad (15)$$

where γ_N is the average labour productivity, which increases at an exogenously fixed growth rate set by parameter ξ_N .

2.5.6. Investment Demand

Based on its expected production plans $\hat{q}_C$, given the Leontief technology, the CGP determines its target level of capital endowment $\bar{K}$ necessary to meet its plan as:

$$\bar{K} = \min(\hat{q}_C, \gamma_N N_{\text{tot}} / \gamma_K) \quad (16)$$

where the minimum level is needed to avoid the CGP undertakes an investment to meet a production target which is not feasible considering the Leontief technology and the limited labour capacity, which is bounded to N_{tot} and present labour productivity. It also worth noting that raw materials are not accounted in the above equation because we assume that the consumption goods firm is never rationed on its imports from the foreign sector.

Given its target level of capital $\bar{K}$, considering its present capital endowment K_{tot} subject to the depreciation rate ξ_K , the firm undertakes an investment I_K given by:

$$I_K = \max(\bar{K} - (1 - \xi_K)K_{\text{tot}}, 0) \quad (17)$$

The choice between investing in green or brown capital goods is made according to the highest Net Present Value (NPV) between green and brown investments expected cash flows, given by:

$$NPV_j = -p_{K_j} I_{K_j} + \sum_{t=1}^{+\infty} \frac{\Delta q_C p_{C,t} - w_C \Delta N_{C,t} - \Delta q_C \alpha_j p_{R,t}}{(1 + r_D)^t} \quad (j = \text{green/brown}) \quad (18)$$

where I_{K_j} represents real investments in new capital goods (green or brown);

p_{K_j} is the present price of capital goods (green or brown);

Δq_C is the additional production (and sale⁹) due to investments;

⁹ The assumption here is that all the additional production amount Δq_C due to investments will be sold.

$p_{C,t}$ is the expected consumption goods sale price at the next t -th simulation step;

r_D is the present loan interest rate on debt, which is based on a mark-up on the Central Bank rate;

w_C is the salary paid to workers in the consumption goods production sector (assumed to stay constant over time and equal to brown sector wage, i.e. $w_C = w_{\text{brown}}$);

$\Delta N_{C,t}$ is the additional amount of workers required at the next t -th simulation step to match the additional production capacity due to investments¹⁰;

$p_{R,t}$ is the expected raw materials price at the next t -th simulation step.

The rationale of the NPV calculation is to compare the present cost of investments, which is higher in the case of green capital goods because $p_{K_{\text{green}}} > p_{K_{\text{brown}}}$, with the present value of related future expected positive and negative cash flows. In particular, we can distinguish three cash flows, namely a positive cash flow, given by the additional sales due to investment, and two negative ones, where the first is given by the additional labour costs, required to match the need for increased production capacity, while the second one is given by the additional raw materials costs incurred to produce the additional output. This latter term is lower in the case of green capital investments, because the relation $\alpha_{\text{green}} < \alpha_{\text{brown}}$ always holds and this difference may more than offset the higher present costs of green capital goods.

Considering the expected cash flows as perpetuities, using the well-known limiting formula for geometric time series, the previous NPV formula can be simplified as it follows:

$$NPV_j = -p_{K_j} I_{K_j} + \frac{\Delta q_C p_C}{r_D - \pi_C^e} - \frac{w_{\text{brown}} \Delta N_C}{(r_D + \xi_N)} - \frac{\alpha_j \Delta q_C p_R}{r_D - \pi_R^e} \quad (j = \text{green/brown}) \quad (19)$$

where p_C and p_R are the present consumption goods and raw materials price, respectively;

π_C^e is the expected inflation in the consumption goods market price;

π_R^e is the expected inflation in the price of raw materials and ξ_N is the labour productivity growth rate.

2.5.7. Credit Market

The money supply of the EIRIN economy is displayed by the level of demand deposits, which gathers the deposits of the consumption and production sectors, as well as government's ones (see Table 1). The maximum credit supply of the bank is set by its equity level E_B divided by the capital adequacy ratio (CAR) parameter, according to the Basel II banking regulatory provisions. The additional credit that the bank can provide at each time step is given by its maximum supply, diminished by the amount of loans already outstanding. Credit is demanded by the consumption goods firm at each time step in order to fully finance the investment demand $p_{K_j} I_{K_j}$ (with $j = \text{green or brown}$) and to cover its negative balances. The new loan ΔLoans granted by the bank to the CGP is set at the minimum between demand and supply:

$$\Delta \text{Loans} = \min\left(\frac{E_B}{\text{CAR}} - \text{Loans} p_{K_j} I_{K_j} + \max(-M_C, 0)\right) \quad (j = \text{green/brown}) \quad (20)$$

Therefore, the consumption goods firm can be rationed in the credit market and, in this case, it scales down its investment plan.

2.5.8. Labour Market

The total labour force N_{tot} is inelastic and supplied by the representative worker. Labour is demanded by the CGP and either by the

¹⁰ It is worth noting that $\Delta N_{C,t}$ decreases exponentially over time due to the increasing labour productivity at the rate ξ_N . We also assume that workers are paid at the brown nominal wage rate in the consumption goods sector and that there will be no rationing in the labour market.

KGP green or brown, depending of the type of capital (green or brown) used by the CGP. In particular, the CGP labour demand N_C^d is determined by its production plan $\hat{q}_C$, by the capital endowment K_{tot} and by the Leontief technology:

$$N_C^d = \min(\hat{q}_C \cdot \gamma_K K_{tot}) / \gamma_N \quad (21)$$

where the minimum is necessary to prevent the firm to hire more labour force than the one required by its productive capacity, which is given by the capital endowment.

Considering that labour is the only productive factor for KGP, the labour demand $N_{K_j}^d$ (j = green/brown) by the KGP (green/brown) that has received a demand for capital goods I_{K_j} (j = green/brown) is given by:

$$N_{K_j}^d = I_{K_j} / \gamma_N \quad (j = \text{brown/green}) \quad (22)$$

The labour transaction N is set at the minimum between demand and supply:

$$N = \min(N_{tot} N_C^d + N_{K_j}^d) \quad (j = \text{green/brown}) \quad (23)$$

In the case of rationing on the demand side, we assume that KGP green and brown have priority over the consumption goods firms. Finally, we assume that the KGP green always hires the skilled fraction skilled fraction of the labour force, consistently with the narrative described in Section 2.2.5.

2.5.9. Production of Green/Brown Capital Goods

Based on the demand of capital goods I_{K_j} (j = green or brown), the KGP produces an amount of capital goods $q_{K_j} \leq I_{K_j}$ based on the hired labour force N_{K_j} :

$$q_{K_j} = \gamma_N N_{K_j} \quad (j = \text{green or brown}) \quad (24)$$

Newly produced capital goods will be delivered to the CGP at the next simulation step.

3. Model Scenarios and Results

We have performed a set of simulations implemented using the software Matlab, each running four different policy scenarios. Two scenarios are characterized by **no government's intervention**, i.e., no public support for green investments. We name them “**Business as Usual**” and “**Coercive Green**”. Then, we introduce two scenarios with **government's intervention in** the form of green fiscal policies and green

sovereign bonds, and we name them respectively “**Green Incentives/Taxes**” and “**Green Incentives/Bonds**”. Below we provide a description of the four scenarios:

- In the “**Business as Usual**” scenario there are no public incentives for investments in green capital goods. The CGP's decision to invest in green or brown capital goods is made according to the highest expected NPV, where brown capital goods are cheaper than the green ones (and thus are often preferred) but they require more raw materials per unit of output. This leads to negative future cash flows due to higher production costs linked to raw materials costs. Therefore, *ceteris paribus*, green capital goods are generally preferred when interest rates are low, i.e., when future cash flows are discounted at a lower rate.
- In the “**Coercive Green**” scenario, there are no public incentives for green investments, but we assume that the CGP always invests in green capital goods (e.g., because forced by government regulations, by ethical considerations about the environment, or in order to meet the changing consumers' preferences).
- In the “**Green Incentives/Taxes**” scenario the CGP always invests in green capital goods but, as a difference from the “**Coercive Green**” case, the government reimburses the producer the difference in NPV¹¹ - if positive - between brown and green investments. The green incentive measure is financed through increases in regular taxation.
- The “**Green Incentives/Bonds**” scenario differs from the “**Green Incentives/Taxes**” because the government covers the difference in NPV between brown and green investments through the issuance of green sovereign bonds, which are bought by the commercial bank. This assumption can be justified considering that green sovereign bonds may represent eligible assets that the commercial bank could sell to the Central Bank in the case a green QE is introduced.

Table 2 shows the values of the key model parameters used in the four scenarios. We assume that the exogenously-given raw material price stays constant and that, consistently, the expected inflation of raw materials prices π_R^e is set to zero. It is worth noting that this choice is conservative. In fact, scenarios characterized by increasing raw materials prices would strengthen the economic benefits of green policies, which would provide comparatively higher savings for the private sector with respect to the case of constant prices.

Concerning the consumption goods price, for the sake of simplicity we assume naïve expectations of constant prices, and thus the expected inflation of consumption goods prices π_C^e is zero. In order to avoid arbitrary choices among a number of algorithms of adaptive expectations, and their impacts on simulation dynamics, we opted for the simplest one, where the price at the next time step is assumed to be equal to the latest observed one.

Figures from 3 to 7 show the results of the simulations of the four scenarios for the real economy and the credit sector. The simulations show the policy differentiation that characterizes the four scenarios from time-step 200, i.e. when the temporal simulation reaches a defined and stable path. The currency unit is indicated with *c.u.* and it is used in relation to monetary variables, in order to distinguish these values from the *a*-dimensional numbers that are used, for instance, in the case of capital endowments, which represent the counting of elementary units (such as in the case of green bonds' holdings).¹²

Each scenario is represented by a different colour and line style. The

Table 2
Values of the key model parameters.

Symbol	Scope	Value
μ_c	Consumption goods price mark-up on unit costs (%)	10
μ_k	Capital goods (green and brown) price mark-up on unit costs (%)	10
α_{green}	Resource intensity of green capital goods	0
α_{brown}	Resource intensity of brown capital goods	0.0015
ξ_N	Labour productivity growth rate (%)	0.30
γ_k	Brown/green capital goods productivity	1
ω_π	Weight of the inflation term	1
ω_u	Weight of the unemployment term	0
$\bar{\pi}$	Inflation target	0
$\bar{u}$	Unemployment target	0
γ_1	Money wage setting parameter (1)	0.018
γ_2	Money wage setting parameter (2)	0.02
w_{min}	Minimum wage	0.50
ρ	Consumption adjustment speed	0.02
φ	Target level of liquid wealth to income ratio	10
ξ_k	Capital depreciation rate	1%
p_r	Constant raw material price	0.50
CAR	Capital adequacy ratio	10%
μ_{bond}	Bond risk premium on Central Bank's interest rate (%)	10

¹¹ It is worth noting that the government pays for the positive difference in NPV, if any, and not for the difference in the present price of capital goods. This is consistent with an investment decision making that accounts not only for the difference between green and brown capital goods prices, which would always favor brown investments, but also for the present value of expected future cash flows.

¹² This means that the value 4 for firm's capital in Figure 3 should not be compared to value 3000 for green bonds' holdings in Figure 4.

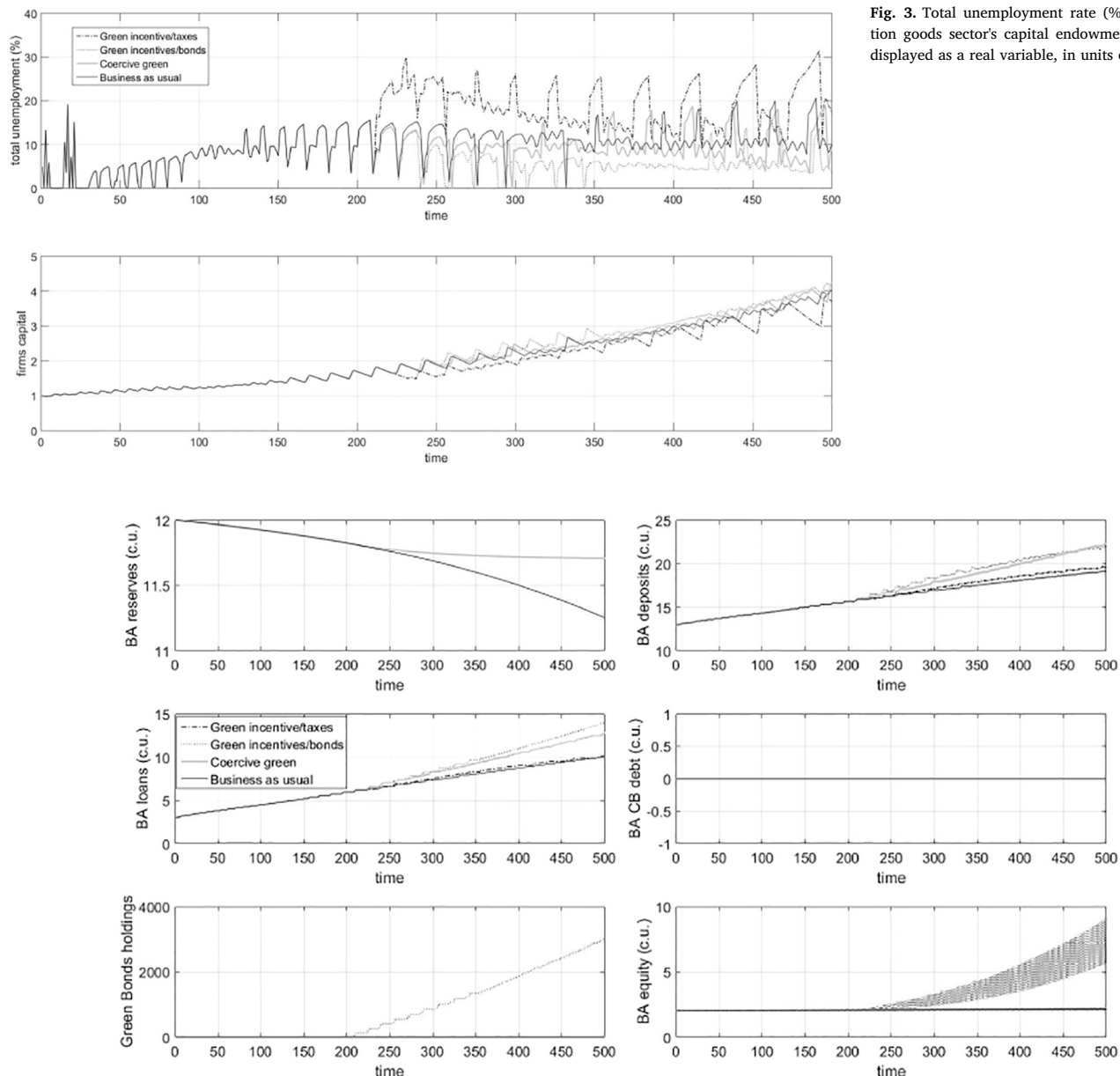


Fig. 3. Total unemployment rate (%) and consumption goods sector's capital endowment. The latter is displayed as a real variable, in units of capital goods.

Fig. 4. Banking sector's (BA) balance sheet. The grey line ("Coercive Green" scenario) in the BA reserves panel (top-left corner), which covers (and thus also represents) both the "Green Incentives/Taxes" and the "Green Incentives/Bonds" scenarios. The black line in the Green bonds holdings (bottom-left corner) and in the equity panel (bottom-right corner) covers the "Green Incentives/Taxes", the "Coercive Green" and the "Business as Usual" scenarios. The other (grey) line, instead, represents the "Green Incentives/Bonds" scenario. The increasing level of equity in that scenario is given by the accumulation by the bank of the payments of the green bonds coupon. All the four cases are represented by the black line in the Central Bank (CB) loans panel (mid-right position), as no borrowing from the Central Bank occurs during the simulations. Bank's deposits keep growing as a consequence of endogenous money creation by bank's lending in all scenarios. With the exception of Green Bonds holdings, which is expressed as units of bond assets, all the other variables are monetary variables and are expressed in currency units (c.u.).

simulations are 500 steps long. Time steps are purely indicative and are not specified in real terms. After a transition period of around 200 steps, the trajectories exhibit a relatively stable and clear behaviour. In particular, if we consider the "Business as Usual" scenario (black line) as the baseline scenario, we clearly see that in the "Coercive Green" scenario the effects of green capital investments on the real economy are positive. This scenario displays lower unemployment and higher capital accumulation on average with respect to the baseline scenario, see Fig. 3. The difference can be explained by the lower import cost of raw materials in the "Coercive Green" scenario that allows for higher purchasing power for households and firms, and thus to better economic dynamics (in terms of consumption, production and employment) with respect to the baseline scenario, where most of capital investments are brown.

Bank's reserves in the "Business as Usual" scenario are well below the level of the other three scenarios, i.e., the "Coercive Green", "Green Incentives/Taxes" and "Green Incentives/Bonds", as a consequence of sustained import of raw materials to support CGP's production, because investments and then capital accumulation are still mostly based on brown capital goods (although small investments in green capital goods are possible). This explains the more than linear decrease of commercial bank's reserves in the "Business as Usual" scenario as a consequence of the growing bill of raw materials import.

On the contrary, in the three green scenarios, bank's reserves decrease at a diminishing pace after time step 200 and become nearly flat at the end of the simulation period, due to the falling costs of raw materials import (see Fig. 5). The reason is that after time step 200, in the three green scenarios, investments are directed only towards green

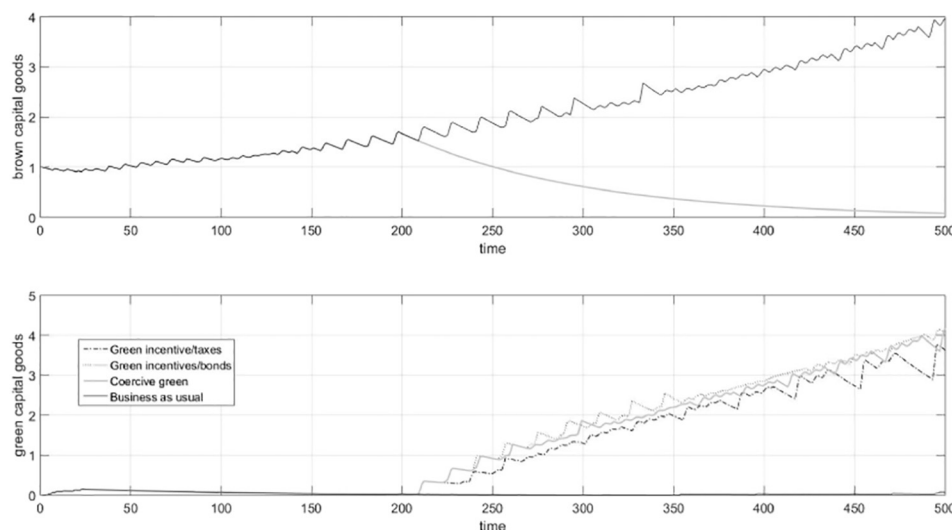


Fig. 5. Green and brown capital goods composition. The grey line (“Coercive Green” scenario) in the brown capital goods panel represents also the “Green Incentives/Taxes” and the “Green Incentives/Bonds” scenarios, as in these three cases all new investments are made in green capital goods after step 200, when the policy differentiation is allowed. With no new brown investments, the brown capital goods endowment depreciates according to an exponential rate. In the case of green capital goods (bottom panel), after step 200, the highest level of investments in green capital goods takes place in the “Green Incentives/Bonds” and the “Coercive Green” scenarios and, at lower levels, in the “Green Incentives/Taxes” scenario. It is worth noting that green investments are not precluded in the “Business as Usual” scenario because firms decide whether to invest in green or brown according to their expected profitability and to how they discount the future (i.e., the NPV). Both variables are displayed as real variables, in units of capital goods.

capital goods, which do not require raw materials for the production. It follows that, in absence of new brown investments, CGP's brown capital endowment as well as raw materials' import decrease at an exponential rate due to capital depreciation.

The reason why the commercial bank's reserves are affected only by the raw material bill is the following. First, as outlined in Table 1, the bank's reserves are composed by its net worth plus the loans received from the Central Bank, plus domestic sectors' deposits, net of loans provided by the commercial bank to the CGP. Second, all the domestic sectors of the EIRIN economy (i.e., the worker, capitalist, CGP, KGP green and brown, and the government), with the exception of the commercial bank itself and the Central Bank, have their accounts in the commercial bank (see Table 1). Therefore, the commercial bank's deposits, i.e., the money supply in the EIRIN economy, are not affected by the transactions among these sectors of the economy. Conversely, commercial bank's deposits are affected by the transactions of these domestic sectors with the foreign sector, i.e., the raw material import cost, and by the capital and current account cash flows with the commercial bank itself. Capital account cash flows are given by the endogenous money creation/destruction that takes place by means of new loans or the repayment of existing loans (see for instance McLeay et al., 2014), while current account flows regard CGP's loans' interest payments to the bank, and the dividends paid by the bank to the capitalist household. In this respect, it is worth noting that if the commercial bank's profit is not fully paid out to the capitalist, then the commercial bank's equity grows at the expense of deposits (i.e., other agents' liquid assets). In conclusion, in absence of Central Bank's lending to the commercial bank, since changes in the commercial bank's net worth and loans are offset by the corresponding changes in sectors' deposits, the variation in bank's reserves depends only on the dynamics of the raw materials import bill.

The two scenarios with government's incentives differ in terms of their effects on sectors' investment behaviours and expectations that depend on the different types of public support introduced, and on the conditions under which they are implemented, i.e., the characteristics of the country's fiscal regime, the presence of budget balance constraints, and the level of interest on sovereign bonds). In the “Green Incentives/Taxes” scenario, unemployment is higher and capital accumulation is lower than in the “Green Incentives/Bonds” scenario; the commercial bank provides less loans, and thus records less new deposits as a consequence of a less performing real economy, which in turn is a consequence of the higher taxation on economic actors. In contrast, the “Green Incentives/Bonds” scenario shows better results for the real economy in terms of unemployment and capital accumulation than in the “Business as Usual” and the “Green Incentives/Taxes” scenarios.

Green sovereign bonds are bought by the commercial bank, and they represent an eligible collateral for the Central Bank in case of green QE. From the one hand, when the bank is the sole legitimate buyer of the green sovereign bonds, their issuance doesn't affect households' purchasing power, with positive effects on consumption, firm's investments (in the green sector, as a consequence of public support) and employment (in green, higher skilled jobs associated to higher wages). Thus, the green economy grows. On the other hand, a word of caution should be spent on the “Green incentives/bonds” scenario. The reason is that the bank receives the interests on the green bonds (coupons), which are paid out by the government, and this solution may negatively affect the economy, in particular in the long-term. In contrast, the increase in households' taxation in the “Green Incentives/Taxes” scenario is immediate. In addition, the accumulation of green assets in the hands of a sole actor, the commercial bank, contributes to increase wealth concentration in the credit sector and in the bank's shareholders (represented by the capitalist household in the EIRIN model), with still unknown long-run effects on financial stability and on social cohesion. We don't focus here on these long-run effects but it is worth remarking that, after the last financial crisis, several analyses showed that wealth concentration in credit and financial institutions did not contribute to economic growth. In contrast, it contributed to accrue inequality (Van der Weide and Milanovic, 2014; Ostry et al., 2014; Hardoon, 2015; Stiglitz, 2015) and to increase the complexity of risk (Battiston et al., 2016a, 2016b). Therefore, in order to avoid unintended – and undesirable effects on system's stability and on inequality – the issuance of green sovereign bonds should be combined with a clear conditionality on the eligibility of investments both for firms (i.e., the green investments in EIRIN) and for the commercial bank, in order to favor bank's lending to the green economy without triggering increases in financial complexity and risk.

Fig. 6 shows the cumulated amount (top panel) and cumulated costs of the raw materials imported in the EIRIN economy, vis a vis the cumulated costs of government subsidies. After step 200, the cumulated amount stabilizes around a constant level in the three green scenarios. In contrast, in the “Business as Usual” scenario (black line in top panel), where the most of the production takes place using brown capital goods, the import of raw materials grows. The reason is that in the three green scenarios, the production capacity of the CGP is increasingly made by green capital goods, which do not require raw materials and thus don't affect import and the commercial bank's reserves. It is worth noting that these cumulated costs of imports correspond exactly to the missing amount in the commercial bank's reserves. The bottom panel shows the cumulated costs of the government's subsidy that contributes to making investments in green capital goods as profitable in terms of

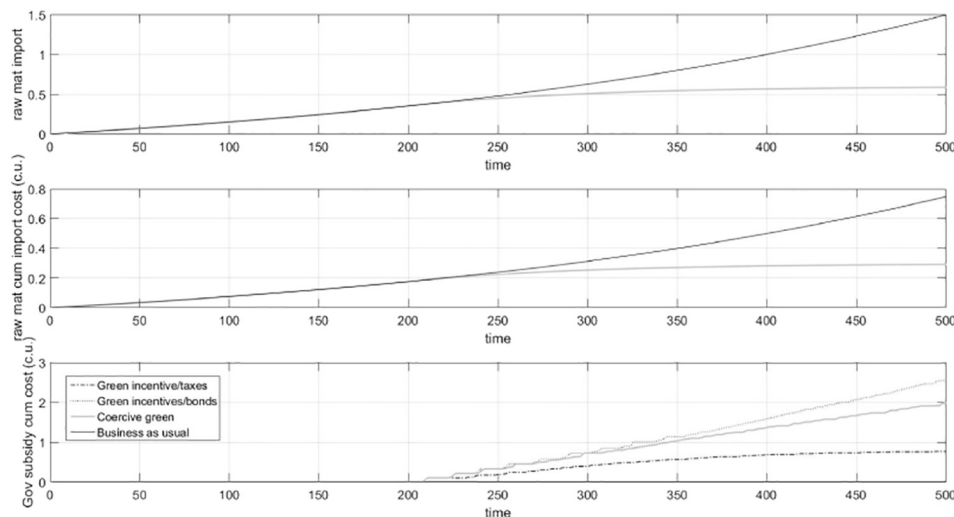


Fig. 6. Cumulated amount (top panel) and cumulated costs (middle panel) of raw materials imports and government's cumulated subsidy costs (bottom panel). The grey line ("Coercive Green" scenario) in the top and middle panels represents also the "Green Incentives/Taxes" and the "Green Incentives/Bonds" scenarios. While the cumulated amount (top panel) is a real variable, both cumulated costs of raw materials imports (middle panel) and government's cumulated subsidy costs (bottom panel) are monetary variables and are expressed in currency units (c.u.).

NPV calculation, as the ones in brown capital goods. In the "Coercive Green" scenario the cost of *going green* (adopting green technology that allows resource resilient processes) is only theoretical and not really faced by the government because it is fully paid by the CGP.

In Fig. 7, the top panel displays the changing dynamics of reserves held by the foreign sector, while the bottom panel shows the sum of reserves held both within the domestic economy and within the foreign sector. Being constant, the cumulated outflow from the commercial bank's reserves perfectly matches in all the four scenarios the inflow of reserves in the foreign sector, due to the cost of raw materials' import.

Finally, the parameter sensitivity analyses show that the high volatility displayed by the unemployment rate is linked to the size of the moving time window - i.e., the memory - which is used by the CGP to compute the average amount of sales occurred in a number of previous time steps equal to the time window size (set equal to 20 in our simulations). This moving average is used as a proxy for the expected sales at the next step, which in turn, given the inventory level, set the CGP's production plan and labour demand (see Eq. (21)). Therefore, the larger the time window size, the lower the unemployment volatility due to the averaging of fluctuations in demand. This is confirmed by Fig. 8 (bottom panel), where the unemployment volatility, computed in the time span between step 200 and step 500, is depicted as a function of the CGP's memory in each of the four scenarios. The ordering of unemployment average among the four different scenarios, computed in the time span between step 200 and step 500 and showed in Fig. 8 (top panel), is independent from the value of CGP's memory, thus confirming the robustness of the model's results.

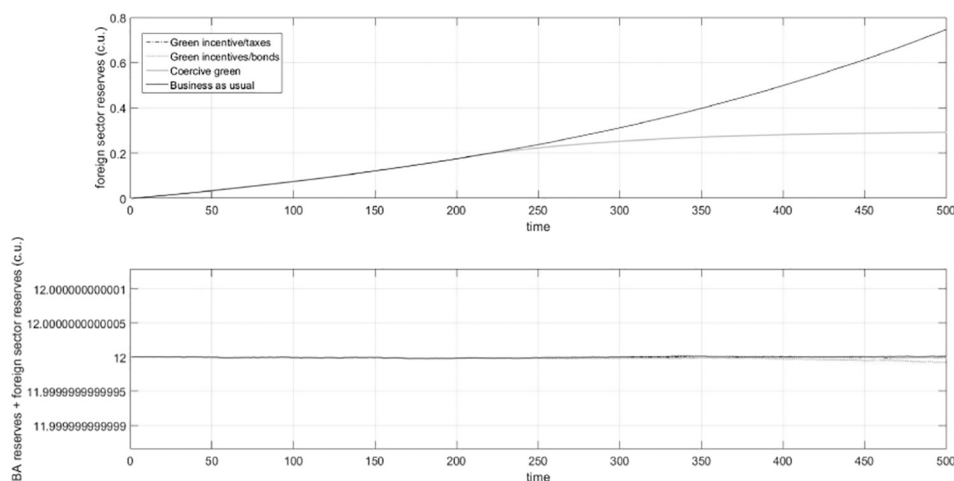


Fig. 7. Foreign sector reserves (top panel) and sum of the domestic bank's reserves and foreign sector reserves (bottom panel). The grey line ("Coercive Green" scenario) in the top panel represents also the "Green Incentives/Taxes" and the "Green Incentives/Bonds" scenarios. In the bottom panel, all four scenarios are represented by the constant black line. Both are monetary variables in currency units (c.u.).

4. Conclusions

We have presented an initial version of the EIRIN model and the results of its application to the analysis of two sets of measures through which governments can support the low-carbon transition, i.e., green fiscal measures or the issuance of green bonds. EIRIN has a peculiar structure that embodies some advantages of SD models, such as the deterministic nature that prevents the risk of over-complexity, and the stocks and flows structure that allows us to identify the causal feedback-loops. At the same time, EIRIN's sectors are modelled as representative agents characterized by their own behavioural functions and adaptive behaviours, and interact through a set of rules. The model's Stock-Flow Consistent structure and the balance sheet accounting approach allow to understand how agents' accounts change in every policy scenario, at every simulation step.

EIRIN endogenizes the linkage between green technology, workers' skills, firms' capital investments, bank's capital requirements, and green policy incentives. With EIRIN we can analyse:

- the impact of green public policy's incentives on firms' divestment decisions from the brown to the green sector (through NPV), and their effects on the real economy and on the credit market;
- the negative effects of production processes that are raw material-intensive on economic growth, domestic employment, balance of payment and bank's reserves, accounting for the irreplaceability of natural resources in the short-term;
- the distributive effects (i.e., wealth concentration and inequality) of

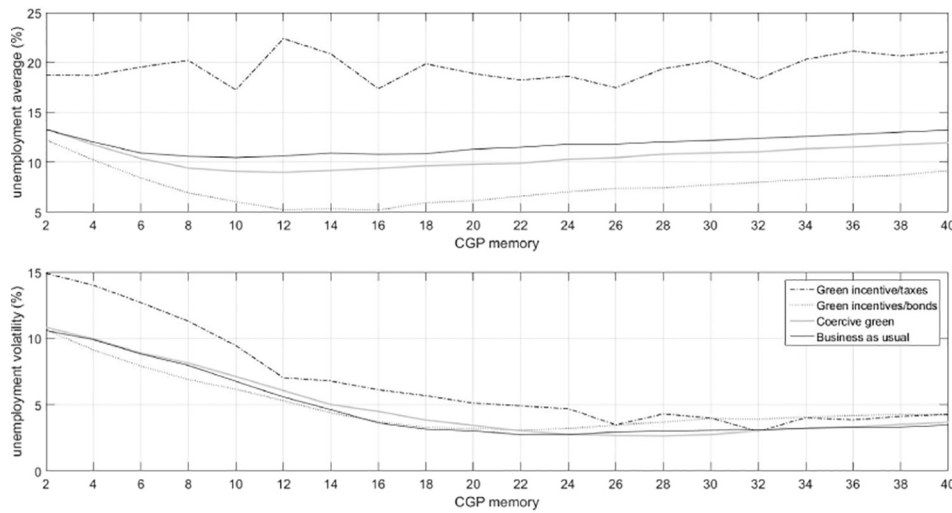


Fig. 8. GCP's memory and unemployment's volatility.

financing the low-carbon transition through green fiscal measures, or through green sovereign bonds, and their effects' time-frame (short-term in the case of the former, long-term in the case of the latter).

The results of the simulations show that, under the model's assumptions, green public policies implemented through the issuance of green sovereign bonds (i.e., the “Green Incentives/Bonds” scenario) represent a win-win solution for the transition to a green economy and the performance of the credit market, thanks to the new green investments that contribute to decrease the import of raw materials. In contrast, the “Business as Usual” scenario, being based on brown investments and production, performs worse in terms of firms' capital accumulation, employment, bank's reserves and deposits, due to the high import of raw materials.

However, green public policies differ in terms of their distributive effects. This difference could be explained by the way and the conditions (i.e., country's fiscal characteristics, budget balance constraints and interest on sovereign bonds) in which they are implemented and financed. The public policies influence households' purchasing power and wealth concentration in the credit sector, with cascade effects on the real economy and on the credit market. Constraining or relaxing the government's budget balance is crucial for explaining the difference between the distributive effects of green fiscal and green monetary policies, in an economy characterized by budget balance constraints (such as the fiscal austerity measures applied in the EU). In the “Green Incentives/Bonds” scenario, where the government is not subject to a zero budget balance, public debt increases as a consequence of the issuance of sovereign green bonds, which in turn has positive spillover effects on green growth and jobs but could negatively affect the worker household in the long-term because of additional public debt burden, thus implying inter-generational equity issues. Since the commercial

bank is the only legitimate actor to purchase green bonds, which represent an eligible asset for the Central Bank in case of green QE, the “Green Incentives/Bonds” scenario is characterized by a trade-off between the positive short-term results for the transition to a green economy and the increase in wealth concentration in the credit sector. In contrast, in the “Green Incentives/Taxes” scenario, the government increases taxation to match the cost of the green policy and meet the budget balance constraint, affecting in particular the worker household's purchasing power already in the short-term. This, in turn, increases unemployment levels and decreases capital accumulation, while the commercial bank provides less loans and opens less deposits as a consequence of a less performing real economy. The next steps of this research will analyse how the trade-offs of the “Green Incentives/Bonds” could be reduced if conditionality on the use of green bonds is introduced by the government.

Acknowledgements

IM thanks Prof. Anthony Janetos of The Frederick S. Pardee Center for the Study of the Longer Range Future for the useful comments and revisions on an earlier version of the draft. IM thanks the IRCRES-CNR for the support to the research at its initial stages. MR thanks The Frederick S. Pardee Center for Global Studies and The Frederick S. Pardee Center for the Study of the Longer Range Future for the financial support during his visit in Boston (USA). IM thanks Stefano Battiston for the suggestion on the presentation of the model's results and their discussion. IM and MR thank Andrea Roventini and Francesco Lamperti for the valuable discussion that improved the quality of the manuscript. IM and MR thanks the two anonymous reviewers for their insightful comments on the model's presentation and results. All the usual disclaimers apply.

Appendix A. Accounting Equations

In the following, we present the accounting equations related to the balance sheet entries, i.e. assets and liabilities, for any representative agent in the model. In particular, we show how assets and liabilities change following the flows of income and expenses, which are determined by the behavioural rules discussed in Section 2.5.

Household – worker:

$$\Delta M_{Hw} = Y_w^{net} - p_C C_{Hw} \quad (A.1)$$

$$\Delta E_{Hw} = \Delta M_{Hw} \quad (A.2)$$

where

$$Y_w^{net} = (1 - \tau)(N_{brown} w_{brown} + N_{green} w_{green}) \quad (A.3)$$

Household - capitalist:

$$\Delta M_{Hk} = Y_k^{net} - p_C C_{Hk} - \Delta n_{brown} p_{bond} \quad (A.4)$$

$$\Delta E_{Hk} = \Delta M_{Hk} + \Delta n_{brown} p_{bond} + n_{brown} \Delta p_{bond} + \Delta(CGP, KGP_s, Bank \text{ share values}) \quad (A.5)$$

where

$$Y_k^{net} = (1 - \tau)(d_C + d_{K_{green}} + d_{K_{brown}} + d_B + n_{brown} c_{bond}) \quad (A.6)$$

Consumption goods producer (CGP):

$$\Delta M_C = p_C q - w_{brown} N_C - p_R q_R - r_D Loans - d_C - p_{K_j} I_{K_j} + \Delta Loans + G \quad (A.7)$$

$$\Delta K_j = -\xi_K K_j + I_{K_j} \quad (j = green/brown) \quad (A.8)$$

$$\Delta I_C = q_C - q \quad (A.9)$$

$$\Delta E_C = \Delta M_C + p_C \Delta I_C + \Delta p_C I_C + \Delta p_{K_{green}} K_{green} + p_{K_{green}} \Delta K_{green} + \Delta p_{K_{brown}} K_{brown} + p_{K_{brown}} \Delta K_{brown} - \Delta Loans \quad (A.10)$$

Green/Brown capital goods firm:

$$\Delta M_{K_j} = p_{K_j} I_{K_j} - w_j N_{K_j} - d_{K_j} \quad (j = green/brown) \quad (A.11)$$

$$\Delta E_{K_j} = \Delta M_{K_j} \quad (j = green/brown) \quad (A.12)$$

Commercial bank:

$$\Delta M_B = r_D Loans + n_{green} c_{bond} - r_{CB} Loans_{CB} - \Delta Loans - \Delta n_{green} p_{bond} + \Delta Loans_{CB} + \Delta Deposits \quad (A.13)$$

$$\Delta Deposits = \Delta M_{Hw} + \Delta M_{Hk} + \Delta M_C + \Delta M_{K_{green}} + \Delta M_{K_{brown}} + \Delta M_G \quad (A.14)$$

$$\Delta E_B = \Delta M_B + \Delta Loans + \Delta n_{green} p_{bond} + n_{green} \Delta p_{bond} - \Delta Deposits - \Delta Loans_{CB} \quad (A.15)$$

Government:

$$\Delta M_G = \tau(Y_w + Y_k) - (n_{green} + n_{brown}) c_{bond} - G + (\Delta n_{green} + \Delta n_{brown}) p_{bond} + Seignorage \quad (A.16)$$

$$\Delta E_G = \Delta M_G - (\Delta n_{green} + \Delta n_{brown}) p_{bond} - (n_{green} + n_{brown}) \Delta p_{bond} \quad (A.17)$$

Central Bank:

$$\Delta M_{CB} = 0 \quad (\text{the value of gold in the vault is assumed constant}) \quad (A.18)$$

$$\Delta Deposits_{CB} = \Delta M_B \quad (A.19)$$

$$\Delta ForeignLiab_{CB} = \Delta M_{ROW} \quad (A.20)$$

$$\Delta E_{CB} = \Delta M_{CB} + \Delta Loans_{CB} - \Delta Deposits_{CB} - \Delta ForeignLiab_{CB} \quad (A.21)$$

Foreign Sector (ROW):

$$\Delta M_{ROW} = p_R q_R \quad (A.22)$$

$$\Delta E_{ROW} = \Delta M_{ROW} \quad (A.23)$$

Appendix B. Wages-Skills Model

Given the average money wage w , which is set by the employment rate according to a Phillips curve-like rule, see Eq. (5), the money wages in the brown and green sectors, i.e. w_{brown} and w_{green} , respectively, are determined according to Eqs. (6) and (7). The rationale behind both equations is to let the aggregate wage bill in the economy, i.e. $(N_{brown} + N_{green})w$, be equal to the sum of the wage bills in the brown and green sector, i.e. $N_{brown}w_{brown} + N_{green}w_{green} = (N_{green} + N_{brown})w$.

In particular, being the labour force characterized by heterogeneous skills, which are uniformly distributed in a continuous range between the lowest level (zero) and the highest value normalized to 1, both Eqs. (6) and (7) are determined by assuming a different wage level associated to each skill, according to a linear relationship, as depicted in Fig. 9. The share of the labour force employed is always the one characterized by the highest available skills. Therefore, wages range between a minimum value, w_{min} , which is associated to the lowest skill among the ones employed in the economy, and a maximum value, w_{max} , which is associated to the highest skill. Given w_{min} , which is a parameter representing a legally set fixed lower bound for wages, and the average money wage w , given by Eq. (5), w_{max} is determined as $w_{max} = 2w - w_{min}$, assuming w to be the midpoint between the minimum and the maximum wages.

Considering that the green sector always employs the share of the labour force characterized by the highest skills, then $w_{green_{min}}$ in Fig. 9 represents the minimum wage in the green sector, as well as the upper bound for wages in the brown sector. Given the relation between w_{min} , w_{max} and w , where w is the midpoint between the minimum and the maximum wages in the economy, then w_{green} can be defined as the midpoint between $w_{green_{min}}$ and w_{max} , and w_{brown} can be set at the midpoint between w_{min} and $w_{green_{min}}$. These definitions lead to Eqs. (6) and (7).

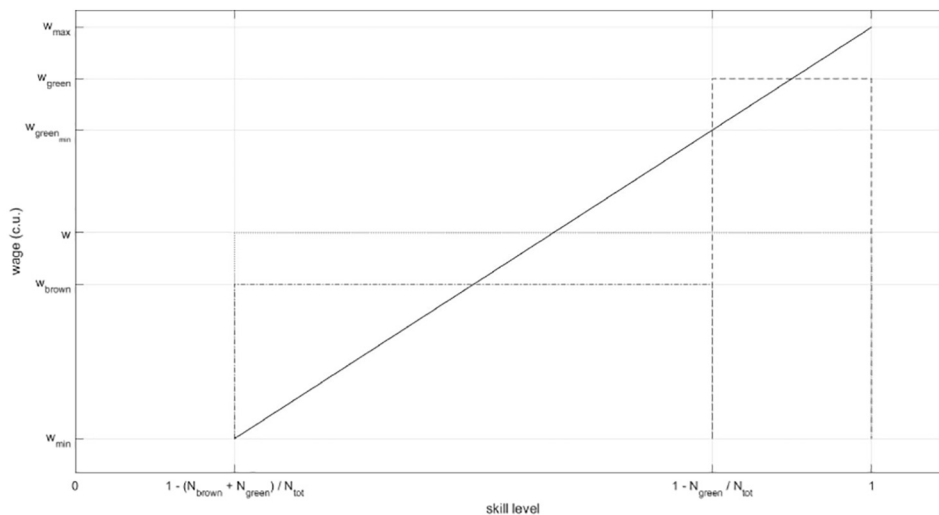


Fig. 9. skills vs wages distribution in the green and brown sectors.

Simple geometric considerations, based on the observation of Fig. 9, confirm that, following the above definitions of w_{brown} and w_{green} , the sum of wage bills in the brown and green sectors is equal to the aggregate wage bill set by the value w . In particular, we can observe that the area included in the rectangle of height w and base $(N_{green} + N_{brown})/N_{tot}$ is equal to the area included within the rectangular trapezium defined by the main relation between wages and skills. This area, in turn, is equal to the sum of the areas of two rectangles, the first one with height w_{brown} and base N_{brown}/N_{tot} and the second one with height w_{green} and base N_{green}/N_{tot} . These areas represent the aggregate wage bill in the economy (or the sum of the wage in the brown and green sectors), scaled by the factor N_{tot} .

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